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Mrittika Roy

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Research Interests

Computer Vision, Medical AI, Trustworthy and Generalizable AI, Multimodal Learning, Explainable AI, Reliable Retrieval-Augmented Systems, Agentic Systems, and AI-Generated Content Detection.

Education

BSc in Computer Science & Engineering Feb. 2019 – May 2024
Rajshahi University of Engineering & Technology (RUET), Rajshahi

- CGPA: **3.66/4.00**
- **Relevant Coursework:** Artificial Intelligence, Neural Networks and Fuzzy Systems, Data Mining, Digital Image Processing, Data Structures, Algorithms, Applied Statistics and Queuing Theory, Database Management.

Publications & Manuscripts

Peer-Reviewed Conference Paper

- Progga Parmita Roy, **Mrittika Roy**, and Mohiuddin Ahmed. "Public Sentiment Analysis of the Israel-Palestine Conflict: A Reddit-Based Study." *2025 International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)*, IEEE, 2025, pp. 1–6. DOI

Manuscript Under Major Revision

- **BGCA-Fusion: Bidirectional Gated Cross-Attention for Multimodal Breast Cancer Diagnosis from Mammograms and Clinical Reports.** Under major revision at *Advanced Intelligent Systems* (Wiley).

Research Experience

BGCA-Fusion: Multimodal Breast Cancer Diagnosis *Manuscript*

- Investigated a limitation in existing breast cancer CAD research, where unimodal analysis and limited fusion strategies can create **diagnostic blind spots** by underutilizing complementary clinical information.
- Developed **BGCA-Fusion**, a multimodal framework based on Bidirectional Gated Cross-Attention to jointly model mammographic images and clinical text through adaptive cross-modal information exchange.
- Conducted evaluation across **three public mammography datasets**, with comparative and ablation analyses indicating the effectiveness of the proposed fusion strategy relative to conventional multimodal fusion approaches.

AI-Generated Image Detection Using Contrastive Learning *Ongoing*

- Investigated the generalization challenge in AI-generated image detection, where detectors may learn dataset- or generator-specific artifacts and consequently struggle to distinguish synthetic images from previously unseen distributions.
- Developed a **contrastive-learning-based detection framework** incorporating frequency-domain representations with supervised contrastive (SupCon) learning, using a real–synthetic image dataset constructed from MSCOCO images and caption-conditioned generated counterparts.
- Conducting **cross-dataset evaluation** to assess robustness beyond the training distribution and investigate whether the learned representations transfer effectively to unseen synthetic-image distributions.

Explainable Dual-Stage RAG for Medical Question Answering *Ongoing*

- Developed a medical Q&A system using Mistral-7B/LLaMA-3 that separates rationale generation from evidence-based answer synthesis.
- Integrated Contriever dense retrieval with BM25, Reciprocal Rank Fusion, and cross-encoder reranking for high-precision medical document retrieval.

Text-to-Image Generation Using GAN Based on XLNet *Undergraduate Thesis*

- Used XLNet-based text representations to extract high-dimensional linguistic features for text-conditioned image generation.
- Implemented DF-GAN to synthesize images through deep fusion of textual and visual feature representations.

Professional Experience

Ztrios Technologies Ltd. - AI Developer Sep. 2025 – Sep. 2026

- Implemented Retrieval-Augmented Generation (RAG) pipelines for AI applications supporting Bengali and performed preprocessing and cleaning of noisy data.

- Fine-tuned Qwen3-4B using parameter-efficient fine-tuning (PEFT/LoRA) with rank 16 and conducted R&D for AI-driven applications.

Samsung R&D Institute Bangladesh - Software Test Intern Jul. 2024 – Jul. 2025

- Developed and executed automated UI test scripts for the Samsung Notes application using Selenium.
- Contributed to preparation of a patent disclosure related to image processing, personalization and privacy.
- Gained conceptual exposure to longitudinal medical data analysis through preliminary research exploration, although empirical work could not be pursued due to the limited availability of suitable medical data.

Selected Technical Projects

Dikkha AI Play Store

- Built a curriculum-aligned Bengali educational Q&A application supporting 20+ NCTB subjects, with subject-aware query rewriting and RAG-based retrieval.
- **Technologies:** Python, FastAPI, Supabase, Google Gemini, Google Cloud Run.

Chhar - Scraper Agent Play Store

- Developed an automated web-scraping agent that discovers retail discount offers, extracts and structures information using LLMs, and publishes enriched data through a backend API.
- **Technologies:** Python, FastAPI, Tavily, Firecrawl, Google Gemini, Google Maps Platform, Google Cloud Storage

System Designer

- Built an agentic platform that collects software requirements, generates system-design documents with Mermaid diagrams, and produces phased implementation plans through a skill-based architecture.
- Developed an LLM router with two-level caching to dynamically select models across Anthropic, OpenAI, and OpenRouter based on task requirements.
- **Technologies:** Python, LangGraph, FastAPI, PostgreSQL, MCP, Docker.

Technical Skills

Programming: Python, C, C++, SQL
Deep Learning & Multimodal AI: PyTorch, Transformers, PEFT/LoRA, CNNs, Vision Transformers, DINOv2
LLMs, RAG & Agentic Systems: LangGraph, dense retrieval, BM25, reciprocal rank fusion, cross-encoder reranking, prompt/workflow design, agentic system design
Retrieval & Data Systems: FAISS, Supabase, document preprocessing, embedding-based similarity search
Backend & Cloud: FastAPI, Google Cloud Run, Google Cloud Storage
Research Tools: Git, GitHub, LaTeX, Overleaf, Kaggle, Google Colab, Jupyter Notebook

Language Skills

Languages: Bengali (Native); English (IELTS: Overall **7.0**: L - 7.5, R - 6.5, W - 7.0, S - 6.5 — Test Taken: Oct. 12, 2026).

Honors & Awards

Dhaka Board HSC Scholarship: Received a general-grade scholarship with full tuition waiver and stipend during Bachelor's studies.
Dhaka Board SSC Scholarship: Received a talent-pool scholarship with full tuition waiver and stipend during Higher Secondary studies.

References

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